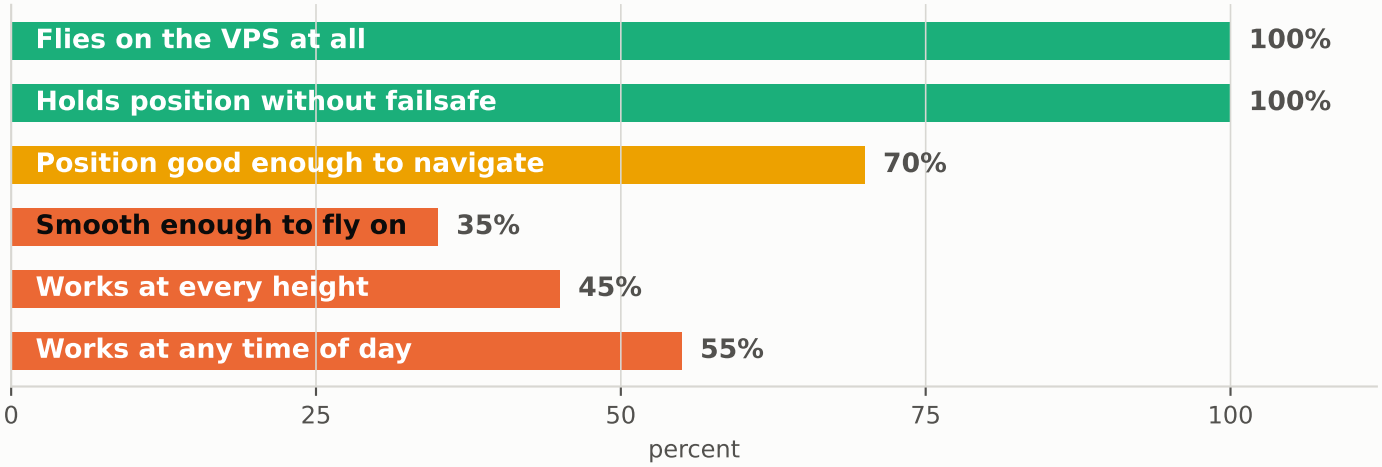
DroneVPS

GNSS-denied navigation on a ZD850 hexa · status, 25 October 2026

branch recover/post-ssd-loss-19oct @ 3959f41 · 61 test files pass

Where the project is



OVERALL: 78%

Up from 62% on 24 October. The jump is one flight: 245.

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THE FLIGHT THAT CHANGED IT  —  245 / vps_00165, 25 October

    634 seconds on the VPS, LOITER throughout, ZERO ALT_HOLD.
    Every flight before it fell to ALT_HOLD two to four times.

        position error vs the raw receiver    0.44 m at 38 m
                                              1.42 m at 82 m (first two minutes)
        map fixes                             159, error p50 5.64 m
                                              2.89 m between 60 and 80 m
        velocity stale                        0.2% of rows

    The system is now MAP-LIMITED, not drift-limited. The map's closing rate is
    0.192 m/s against a far slower drift, so the position settles at roughly the
    map's own accuracy. That is a different regime from every earlier flight.


WHAT GOT IT THERE, in order of size

    Per-track derotation      A rotation moves a point at radius r by
                              f·theta·(1 + r²/f²). The code subtracted ONE
                              centre-valued shift from a 77.7° frame, leaving
                              +25.6% of the rotation behind. Predicted 25.6%,
                              measured 16-32%. This was the phantom.

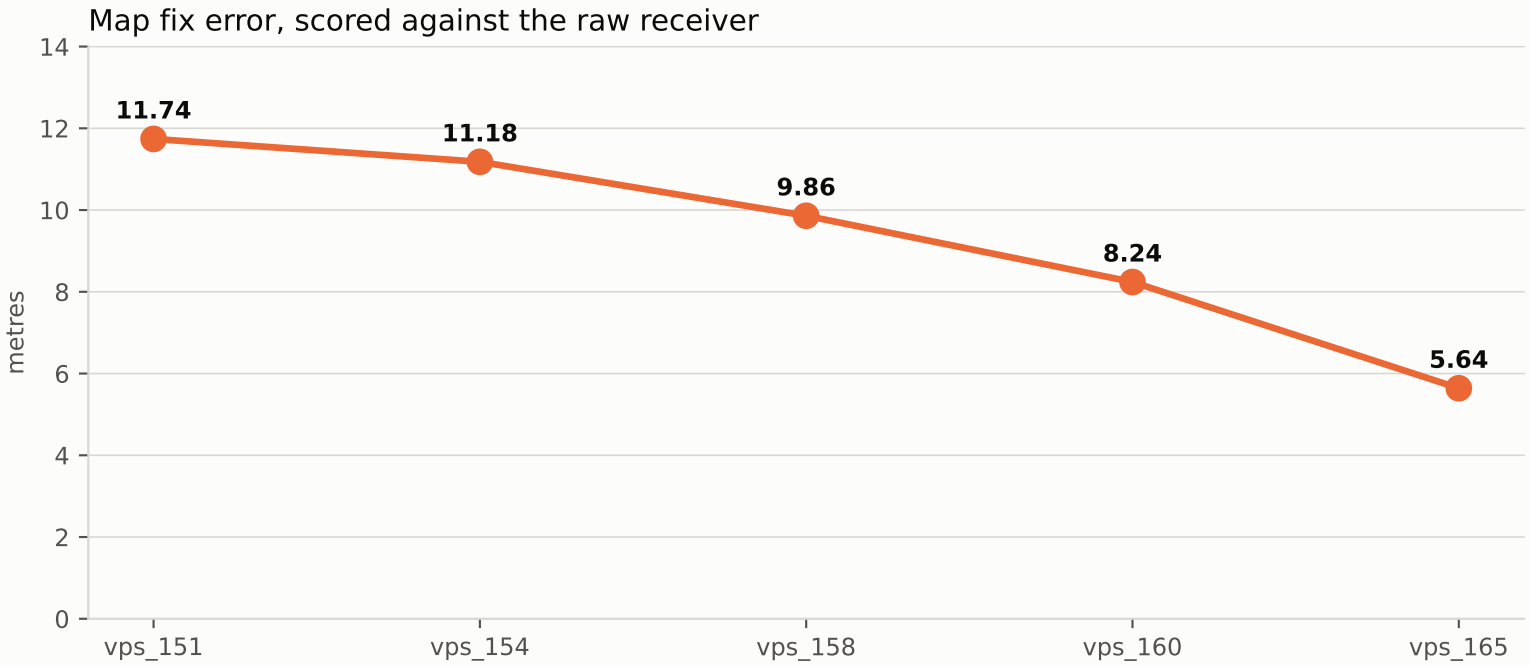
    The tilt correction       A tilted camera's image centre is alt·tan(tilt)
                              from nadir, and the matcher reported that centre
                              as the aircraft. Map error 11.18 -> 8.69 m.

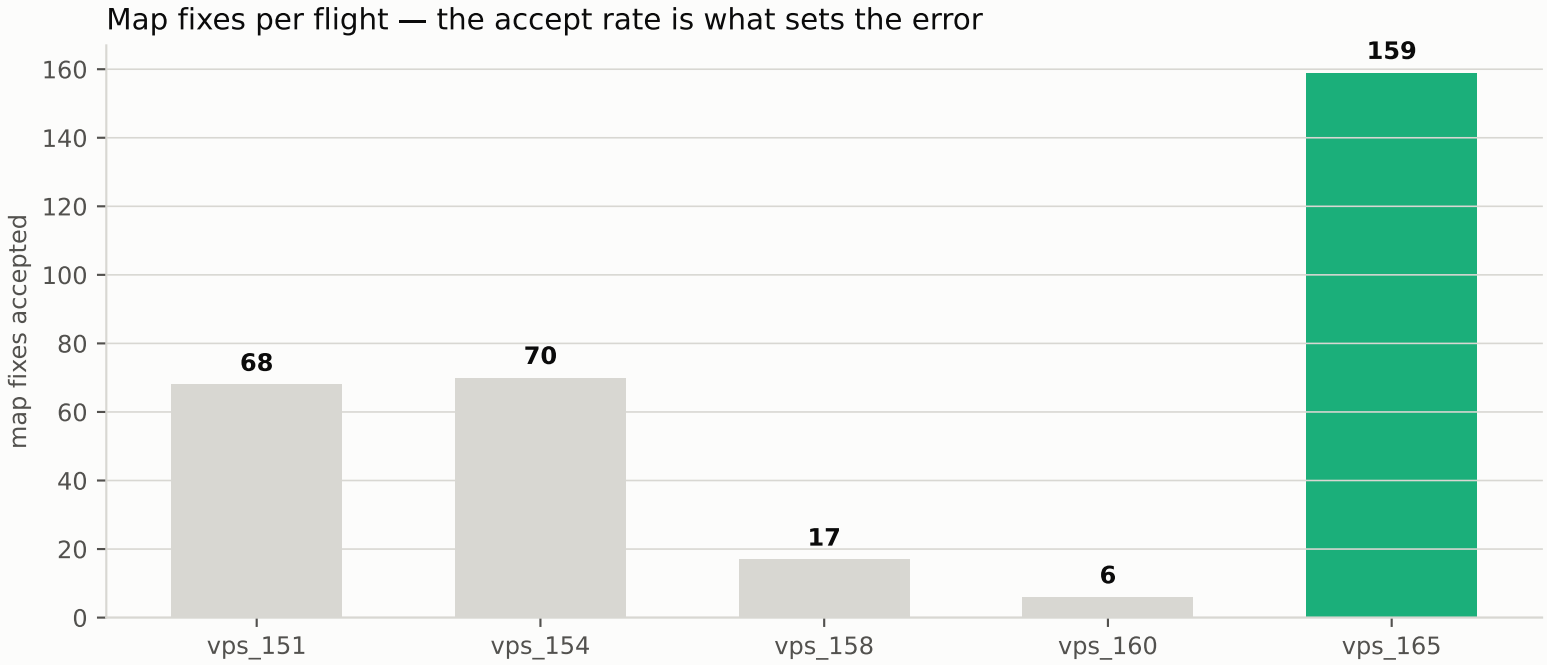
    Both matchers             NCC came back. XFeat accepted 78% of frames at
                              11:38 and 0 of 40 at 15:20 — the shadows had
                              swung 100° from the map's. NCC does not care.

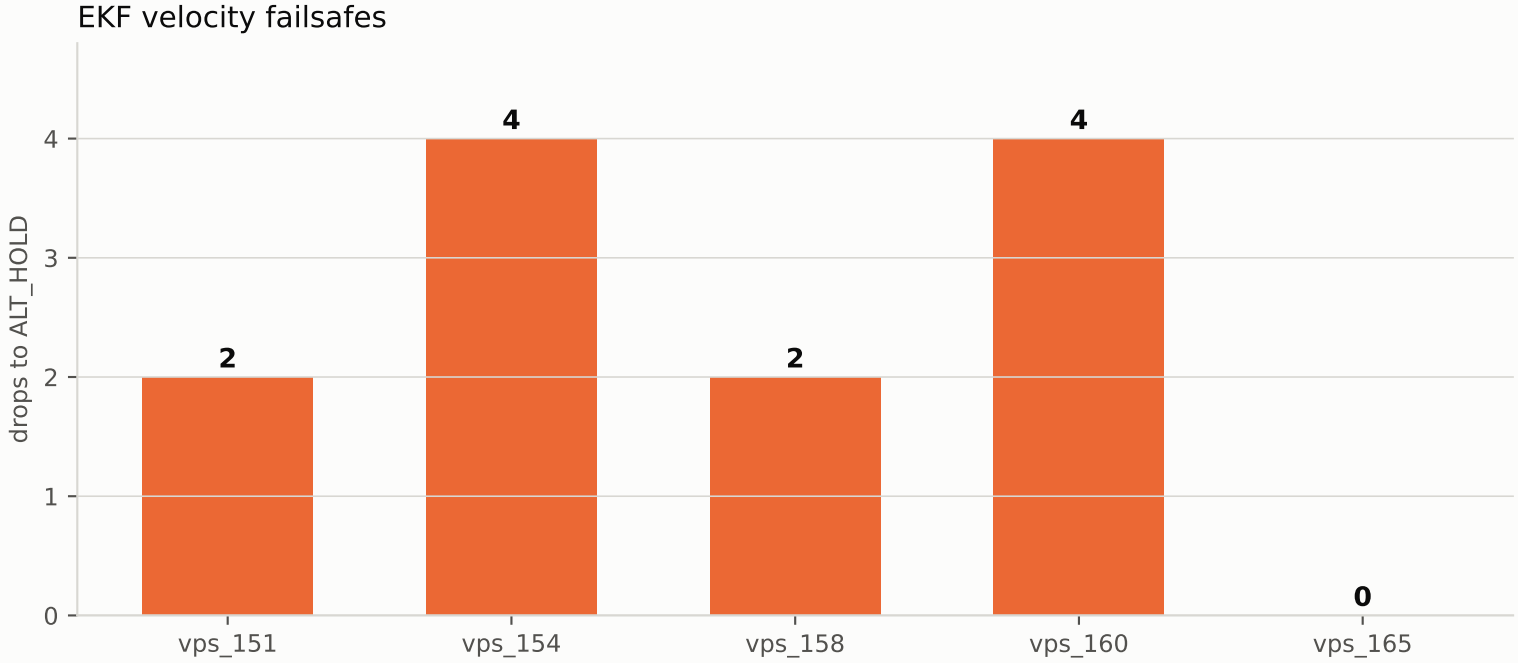
    The fix latency           The map corrected a position the aircraft had
                              already left. The prior was carried in the
                              result and never read.

    Camera yaw offset         -1.95°, measured on two flights.
```

What the numbers did







What is still wrong

1. THE AIRCRAFT JERKS — OPEN

The sent speed crosses zero 307 times in 634 s — 0.48 a second — and the deadband zeroes 39.9% of rows. It is a hard threshold: below it the flight controller is told exactly 0, above it the full value, so every crossing is a step. Commanded tilt saturates at ATC_ANGLE_MAX 15°, rates to 56.9°/s. Map fixes are ruled out: 16% coincidence against a 15% base rate.

It only became the limiting fault because the phantom finally shrank. It used to sit well above the threshold; now it sits ON it.

2. TOILET BOWL AT 35 m — OPEN

The bearing from its own centre swept 754° in 24 s while the yaw moved 9.6° — circling, not turning. Period 11.4 s, and 2·pi × the 40-frame baseline's 1.6 s lag is 10.1 s. The baseline's delay IS the period.

3. THE BAROMETER IS 5-7 m HIGH — OPEN

Against the lidar: 1.07× at 100 m 1.29× at 20 m
1.17× at 40 m 1.72× at 10 m

The same climbing and descending, so an offset and not hysteresis. At touchdown the lidar read 0.60 m and the altitude in use read 4.2. Velocity is pixels × altitude / focal, so that ratio IS the velocity error — 72% at 10 m, which is where the aircraft was circling.

4. THE MAP ABOVE 95 m — OPEN

11.37 m against 2.89 m between 60 and 80. Fly the map work low.

5. ERROR CREEPS — OPEN

1.4 m to about 10 m at 82 m over a few minutes.

WRITTEN FOR 1, 2, 3 AND 5 — THEN REVERTED, UNPROVEN

5066572	the deadband as a ramp
15d712f	two comments that wrongly said the lidar carries the height
e626f42	flow_vel_frames 40->30, fix_sigma_m 20->8, lidar correcting the barometer's offset without replacing it

The pilot flew once more with the origin set by hand from SCR_USER1/2 rather than from the receiver. It tracked worse. That FC log is deleted and the tree is back to exactly what flight 245 flew.

The confound is real: the hand-set origin and three code changes went out together, and nobody knows which caused it. REINTRODUCE THEM ONE AT A TIME.

CLOSED BY MEASUREMENT, DO NOT REOPEN

the velocity baseline length	flat from N=20 to 160 before the derotation fix; worth revisiting now
scale / focal	0.79-1.02 against truth
the camera accelerometer	never free-runs, p99 age 0.18 s
the gyro integration window	bit-identical on 6400 frames
tracker noise	98% of the step is common to both cameras, so it is real image motion
the camera-to-FC lever arm	0.157 m/s at 30°/s, against a 1.4-2.5 m/s phantom
raising max_correction_m	flown 3 Oct, hold fell 11-14 s to 4.5-9.0 s
a height-scaled velocity baseline	flown log 229, ran 11.1 m in 24 s